

Resolving the Remaining Cases of Carlitz-Extensiveness over Finite Fields using SageMath

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Abstract. The primitive normal basis theorem asserts the existence of elements in a finite extension of finite fields which simultaneously generate the multiplicative group and a normal basis over the base field (Lenstra-Schoof, [1]). This work deals with a generalisation of this problem.

Hsu and Nan [2] proposed a Carlitz-module analogue of this theorem, replacing primitive normal elements by primitive generators for finite Carlitz modules. Their work establishes the desired existence result with one confirmed exception, $(q, n) = (2, 2)$, and a finite, effectively computable list of further possible exceptional cases. Hachenberger [3] subsequently placed this question into the broader framework of primitive generators for cyclic vector spaces over Galois fields; in particular, his work resolved 41 of the 63 undecided pairs left by Hsu and Nan. In the concluding remarks of Hachenberger’s paper, computations by Gruber using SageMath further reduce the Carlitz-extensiveness problem to six remaining undecided pairs. The present work resolves these remaining cases.

The talk explains the computational part of this final step. We present a fully reproducible SageMath implementation designed to handle the six undecided parameter pairs left after the reductions of Hsu–Nan, Hachenberger, and Gruber. The implementation combines finite-field techniques, including computations with cyclotomic cosets, with projective-space methods and minimal-polynomial-based verification procedures for the relevant generator properties.

Together, these computational results establish the Carlitz-module analogue of the primitive normal basis theorem apart from the single exceptional pair $(q, n) = (2, 2)$ identified by Hsu and Nan. In particular, the resolution of the six remaining pairs confirms that primitive Carlitz generators exist for all other finite field extensions. The code and verification data are intended to be made publicly available.

References

- [1] H. W. Lenstra and R. J. Schoof. “Primitive normal bases for finite fields”. In: *Mathematics of Computation* 48.177 (1987), pp. 217–231. ISSN: 0025-5718, 1088-6842. DOI: 10.1090/S0025-5718-1987-0866111-3. URL: <https://www.ams.org/mcom/1987-48-177/S0025-5718-1987-0866111-3/>.
- [2] Chih-Nung Hsu and Ting-Ting Nan. “A generalization of the primitive normal basis theorem”. In: *Journal of Number Theory* 131.1 (Jan. 2011), pp. 146–157. ISSN: 0022-314X. DOI: 10.1016/j.jnt.2010.05.011. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0022314X10001708>.

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- [3] Dirk Hachenberger. “Primitive generators for cyclic vector spaces over a Galois field”. In: *Australasian Journal of Combinatorics* 64.2 (2016), pp. 289–326. ISSN: 2202-3518. URL: https://ajc.maths.uq.edu.au/pdf/64/ajc_v64_p289.pdf.